

Project Svarog 2026-27 Introductory Project:

Flywheel Control Demonstrator

Aditya Shankar, Nishant Kidangan Mathew, Sahil Bhargava

Motivation 1
Project Objective 2
Timeline 3
Expectations 3
Team Structure 4
Background 4
Technical Specifications 5
  Electrical: 5
  Software: 6
  Mechanical: 6
Useful reading: 6
Good luck! 6

Motivation

After two launches on the European Space Agency's Balloon Experiments for University Students (BEXUS) programme which successfully demonstrated the capabilities of our 8U solar sail CubeSat demonstrator, the 2026-27 academic year for Project Svarog aims to further develop the required electronics, mechanical, and avionics subsystems to get them closer to Low Earth Orbit (LEO) capabilities.

One of the crucial components for us to develop is our Attitude Determination and Control System (ADCS). This is the component which is responsible for controlling the satellite's orientation in space. From the name itself, there are two distinct roles of the component, namely:

1) Determining attitude or the angular position in space
2) Controlling the attitude to a desired position

The first of the two is generally done using a suite of sensors, eg: star trackers, IMU, GPS, but our focus is the second objective: to be able to control the attitude to a desired setpoint.

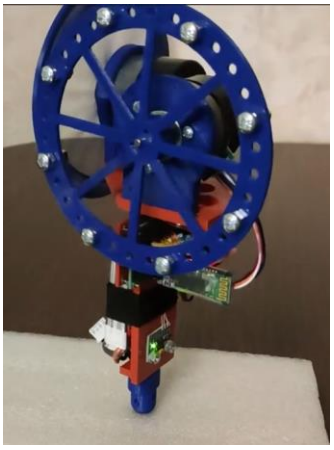
Traditionally, ADCS is done using flywheels which can be spun and despun to impart some angular momentum into the satellite, allowing its orientation to be changed. For this project, we will also be using flywheels as they are well documented with flight heritage (this is an important factor in any space mission design!).

Project Objective

To build a 2-axis flywheel control system which:

1) Fits within 1U (10x10x10cm) size which houses
    a. 2 flywheels (1 per axis)
    b. All required motors and electronics
    c. Attachment to pinpoint stand
2) Will be demonstrated on a pinpoint stand we will provide such that attitude targets can be set, and the demonstrator follows

We have provided an open-source reference below that you can use for inspiration. Although keep in mind that your moments of inertia, and mass distributions will be wildly different (hint: Fusion has a very nifty tool that will give these to you). You will have to take into account how these will affect your equations used, and what the best way to implement them is!

	Link to YouTube Video

Timeline

Date	Event	Description
19-10-26	1st meeting	The Introductory project begins. You will form the group, understand tasks and group member's experience and expertise. The workshops for CAD, PCB design and PID control will begin here.
26-10-26	2nd meeting	Background research should be done, and tasks should be delegated amongst members. Short discussion with team leads post workshop.
2-11-26	Subsystem Leads proposal	Group will choose 3 representatives for electronics, mechanics, software
30-11-26	PDR	Preliminary design review. Individual subsystems (eg: frame, motors, software) to be in good shape (80% done), but low/no integration work started.
07-12-26	Final Autumn term meeting	Design documentation from each subteam (2-3 pg doc) submitted along with materials that need to be ordered following PDR feedback.
11-01-27	Post-winter break status update	Subsystem leads (primarily) discuss feedback from PDR and roadmap for next weeks (important!)
01-02-27	Testing week	Assembly, integration and testing week. See how well the control systems perform.
08-02-27	Project conclusion	Work is finished on the project. It is time to showcase the project to everyone on the team and discuss how it works.
19-03-27	Project report	Submit a report of the project, see BEXUS36 SEDv5 for example. Also, short presentation about the project on this weekly meeting.
19-06-27	Showcase	Opportunity to showcase the project at the Great Exhibition Road festival at Imperial.

Expectations

This intro project is being proposed not only to build expertise in ADCS within the team, but also to make new joiners familiar with the project development lifecycle. This is not an easy project! The deadlines are ambitious and you may run into delays, but it will be highly rewarding at the end!

We expect that you will work in a fairly large group to complete the project. You have been divided into subteams for electronics, mechanical and software for categorization of work but you are expected to work together across subteams. This will be especially crucial when you will need values from different teams and will have to work with each other's requirements to make a design decision.

No prior experience is required and we will teach you the basic knowledge required to start the project. The project leads are always available to help and give guidance if you are stuck. As leads we want to foster a high level of skill within the teams, so we are more than eager to help if needed. We do, however, expect a decent level of commitment to the project. The Monday sessions will serve as checkpoints, and opportunities to discuss design decisions with the leads and consult them. Majority of the design work would be done outside Monday sessions.

Each subteam will have their own WhatsApp channel with the team leads along with a general group for the introductory project. We expect good project management etiquette with documentation. research material, design and code files named and stored approximately.

Documentation	Code	CAD	Communication
Google Drive	GitHub	Fusion	WhatsApp

Throughout the project you can 3D print prototypes out and order other components like standard hardware, PCB's or machined components. However, any expense will have to be justified. At the end of the project, we will be assembling all your designs, and stress testing them! Make sure all your designs are made with DfAM in mind.

Testing and scoring will be based on a mix of CAD and Coding etiquette and efficiency, RMSE of control system along with total Energy draw. **A proper rubric for the PDR and the testing will be shared in the near future! In general, the timeline is open to changes given feedback from the team.**

Team Structure

After two weeks of the intro project running, we will ask for the group to decide on electronics, mechanical, and software leads. These will be the people responsible for

communicating to each of the other members in the respective departments and will be the primary contact point for the leads (obviously we will still be talking to each of you, but we expect leads to be the ones generally knowledgeable).

Background

It is suggested to read up on basic control theory, for example how mechanical systems are represented, as well as feedback control systems and how they work, as well as using multi-loop systems. Understanding practically how PID (proportional integral derivative) control works is also important.

We will be going over how to make bodies and assemblies in CAD using Fusion 360, through workshops hosted by our STRUM leads (Aditya & Thomas), pay close attention to these, where we will share useful tips for assemblies. A good playlist you can go through in your own time is linked here.

For those working with electronics, knowing to make PCBs is important and I would encourage those responsible to attend the **introductory** electronics course which will be hosted by electronics lead (Nishant Kidangan Mathew) and co-lead (Ariya Narayanasamy). For those very eager to start learning, you can take a look at the tutorial series we created last year for Altium. This year we are not using Altium but the flows are similar.

Understanding the operation of brushless DC (BLDC) motors and how BLDC motor drivers work can also be useful for those working with the motors directly. A deep understanding is not required for everyone but it is worth watching this video to get an overview.

Technical Specifications

Electrical:

1) The system can be controlled via SPI (as an SPI slave)
2) The system will use an STM32 microcontroller as its main computer
3) When mounted onto the pinpoint stand, it will be difficult to send signals over wires to the demonstrator, thus an ESP32 (as it has builtin wifi functionality) must also be included to allow the system to be controlled over wifi.
4) The system will use BLDC motors.
5) The motor driver(s) can be either off the shelf or custom made
6) The whole system should be made primarily with custom PCBs to save space.
7) The bottom 10x10x3xcm should be allocated to the battery power system (this will be designed by Nishant and Ariya and will be provided to you) - this space is

also somewhat flexible but we need to know if this space allocation does not suit you latest a week before PDR.

8) Must include a gyroscope/IMU for attitude determination. Other sensors can also be added for better accuracy

Software:

1) All software should be written in either C++ or Python
2) Must implement the control loop by reading sensor data and actuating motors as needed
3) Must include SPI commands for setting the attitude
4) Must include ability to read out sensor data
5) Must be able to determine and report the attitude at any point
6) Must implement a simple simulation to test your control system, record RMSE and tune PID as appropriate. (Can be done in Simulink).
7) Stretch goal: Implement a GUI to control and view the system.
8) Stretch goal: Implement Bayesian tuning / Genetic Algorithms to tune PID gains
9) Stretch goal: Investigate Monte Carlo simulations for your control system (Highly recommend Simulink).

Mechanical:

1) The CAD must be modular as far as possible, with good assembly and sketch etiquette.
2) Parametrise CAD as much as possible. Must have atleast 3 useful parameters implemented.
3) Must follow rules for DfAM. No overhangs greater than 45deg, in direction of printing.
4) Care should be taken to ensure the structure can survive in the case of the errors in the control system leading to uncontrolled falls

Useful reading:

https://commons.erau.edu/cgi/viewcontent.cgi?article=1138&context=beyond

https://www.sciencedirect.com/science/article/abs/pii/S0094576503003382

https://www.youtube.com/watch?v=wkfEZmsQqiA&list=PLn8PRpmsu08pQBgjxYFXSsODEF3Jqmm-y

Use Imperial institutional login for any paywalled journals

Good luck!